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METHODOLOGY AND DATA DESCRIPTION SUBMISSION

Assignment 2

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CHAPTER 3 – ANALYSIS

3.1 Introduction

The methodology used in the development of the Lightweight CNN-Based Food Freshness Detection System Using Smartphone Images for Sri Lankan Market Environments is discussed in this chapter. The methodology comprises of feasibility study, fact-finding approach, requirement analysis, functional and non-functional requirements, development method and evaluation process.

The proposed research work will develop a computer vision-based intelligent image classification system to detect the freshness level of fruit and vegetable items from smartphone images in Sri Lankan market environment condition. While the majority of the current works in literature have been done in lab conditions, the research focuses on realistic market environment where there will be variations in lighting, background and camera quality of smartphones.

3.2 Feasibility Study

A feasibility study is carried out to ascertain whether the research can be implemented.

Technical Feasibility

The implementation of the proposed system will be done using Python, TensorFlow, Keras, and OpenCV. The lightweight CNN models, namely MobileNetV2 and EfficientNetB0, will be implemented through transfer learning, while the traditional CNN model will be created as a baseline model. Data preprocessing and augmentation will help in improving the quality of the data set before training.

Economic Feasibility

The proposed research employs open source software packages such as Python, TensorFlow, Keras, OpenCV, and Google Colab, making development costs minimal. Smartphones will be used to capture the data set, thus avoiding the cost of sophisticated imaging equipment.

Feasibility of Operation

The suggested system has been developed for consumers, food sellers, and food inspectors, who need an easy and cheap way to measure food freshness. The use of smartphones in Sri Lanka has become ubiquitous; therefore, this system does not require any special equipment to operate.

Feasibility in Terms of Law and Ethics

All pictures will be gathered with the consent of vendors or marketplaces following all ethical research practices. The dataset will only include images of fruits and vegetables without gathering any private data.

3.3 Techniques for Fact-Finding

These techniques have been used for collecting facts and system requirements.

1. Literature Survey (Primary Technique)

Latest journals, papers, and prior research work on CNN-based food freshness detection, transfer learning, image classification, and computer vision were reviewed.

As found through literature survey, it is noted that:

Most of the available systems are based on laboratory datasets.

There is lack of study where images from smartphones captured in real market conditions have been analyzed.

Lightweight CNN models provide accurate results without needing high computation power.

2. Observation

Observation of local vegetable and fruit markets was carried out for determining environmental conditions like lighting, backgrounds, etc.

3. Data Set Creation

An image data set is going to be developed through capturing smartphone images of fresh produce of fruits and vegetables available in Sri Lanka.

These will be classified as:

- Fresh
- Medium Fresh
- Rotten

4. Documentation Review

Prior research papers, CNN tutorial guides, TensorFlow documentation, and image processing have been studied to find out the best methodology for the project.

3.4 Requirement Analysis

Requirements have been determined from the literature study and research objectives.

- Stakeholders
- Consumers
- Market Vendors
- Food Inspectors
- Researchers
- Agricultural Authorities
- Requirement from Research

According to the literature study, there is a requirement of:

- Proper classification of food freshness.
- Lightweight CNN models that can run on smartphones.
- Classification under varied lighting situations.
- Model training with minimal computational effort.
- Comparison between traditional CNN and transfer learning model.

3.5 Functional Requirements

The system shall:

- Be able to capture or import images from the smartphone.
- Preprocess the images before training.
- Perform image augmentation.
- Train a traditional CNN model.
- Perform transfer learning for MobileNetV2 and EfficientNetB0.
- Classify food freshness into three categories such as Fresh, Medium Fresh and Rotten.
- Predict food freshness category for new images.
- Compare model performances.
- Generate evaluation report.

3.6 Non-Functional Requirements

- Performance: High classification accuracy with low inference time.
- Reliability: Consistent prediction in various environmental settings.
- Usability: Ease-of-use image classification process.
- Scalability: Capacity to introduce other food types in future work.

- Maintainability: Simplicity in re-training with new data.
- Compatibility: Compatibility with images taken by different Android smartphones cameras.

3.7 Development Approach (Agile)

Agile approach is chosen due to iterative nature and possibility of continuous model improvement.

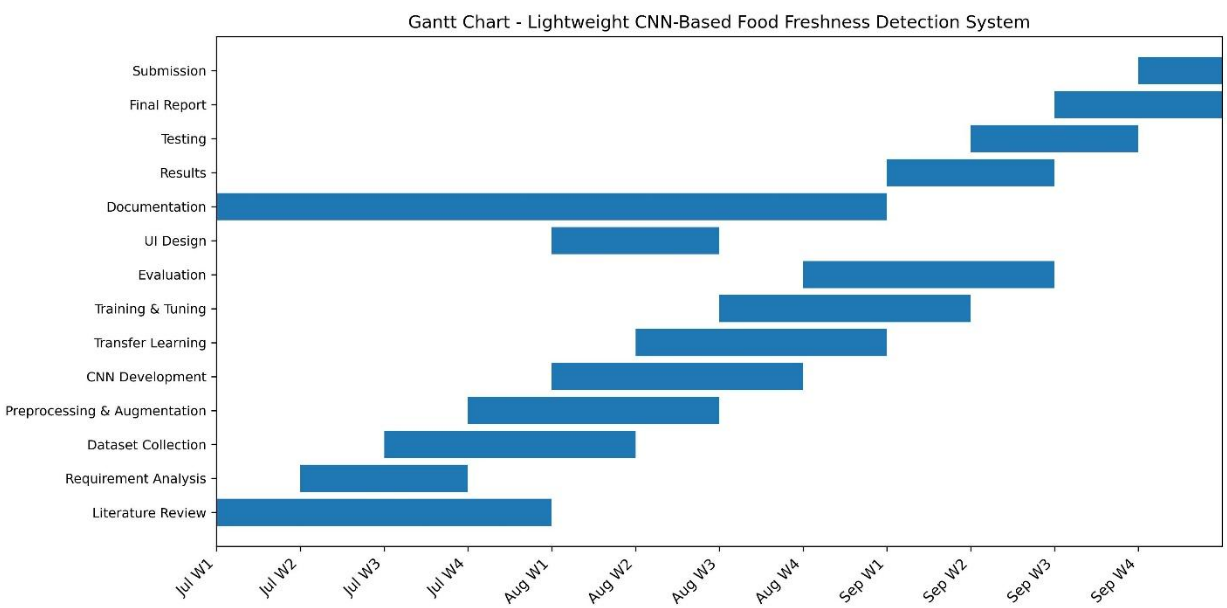
Why Agile?

- Ability for continuous growth of datasets.
- Repeating of experiments.
- Comparison of models.
- Improvement of model performance gradually.
- Phases of Agile Used

Planning

- Definition of research goals.
- Defining food categories.
- Selection of CNN models.
- Defining evaluation metrics.

Gant Chart



Designing

- Design of machine learning workflow.
- Designing of pre-processing technique.
- Model comparison design.

Development

- Collection of smartphone images.
- Preprocessing images.
- Augmentation.
- Training of Traditional CNN.
- Fine-tuning MobileNetV2.
- Fine-tuning EfficientNetB0.
- Evaluating Models

Accuracy

- Precision
- Recall
- F1-Score
- Confusion Matrix

Deployment

Store the trained models to be used for offline prediction and future integration into mobile applications.

Reviewing

Keep improving the models by tuning hyperparameters and evaluating their performance.

3.8 Dataset Collection Methodology

The dataset collection technique is the method that will be adopted to gather the images for training, validation, and testing the proposed model for food freshness detection. As the available publicly available datasets fail to accurately represent real-life situations in Sri Lankan markets, it is necessary to create an appropriate dataset for the current research.

The pictures of selected fruits and vegetables will be collected by the means of cameras installed in Android smartphones from local vegetable markets, supermarkets, and households all over Sri

Lanka. Images will be taken under various lighting conditions, perspectives, distances, and backgrounds to better mimic realistic situations. Such procedure allows increasing the diversity of the dataset and improving the generalization capabilities of the model.

Each image will be carefully examined and annotated according to one of the three freshness states, namely Fresh, Medium Fresh, and Rotten. Labeling will depend on the visual characteristics such as color variations, texture, bruise appearance, mold growth, and surface damage. Then, the labeled dataset will be split into three parts: training, validation, and testing.

To follow ethical standards of research, only the images of fruits and vegetables will be collected and nothing else will be included in the datase

3.9 Dataset Summary

Attribute	**Description**
Dataset Type	Custom Image Dataset
Image	Source Local Sri Lankan markets, supermarkets, and home environments
Image Capture Device	Android Smartphone Camera
Food Categories	Tomato, Banana, Mango, Carrot, Cucumber, Brinjal (Expandable)
Freshness Classes	Fresh, Medium Fresh, Rotten
Image Format	JPG
Image Resolution	Resized to 224 × 224 pixels
Expected Dataset Size	Approximately 1800 Images
Dataset	Split 70% Training, 15% Validation, 15% Testing

CHAPTER 4 – DESIGN

4.1 Introduction

The chapter will introduce the complete design of the Lightweight CNN-Based Food Freshness Detection System Using Smartphone Images for Sri Lankan Market Environments. It contains the design process of transforming the requirements discussed in the previous chapter into a practical deep learning solution. In this chapter, system design, dataset design, data preprocessing, CNN model design, interface design, evaluation design, and hardware/software requirements will be discussed.

The objective of this chapter is to show how the design decisions are contributing towards developing an accurate, lightweight, efficient, and reliable food freshness detection system for Sri Lankan market environments.

4.2 System Design Process

The system design process gives a structured approach that converts the research requirements into a machine learning solution able to classify food freshness based on smartphone images.

The main steps are as follows:

- Requirements Analysis
- System Design
- Component Design
- Interface Design
- Validation Planning

4.2.1 Requirement Interpretation

Functional and Non-Functional Requirements

The requirements analysis involved identifying the needed components and steps to classify the freshness level of foods.

The proposed system includes:

- Smartphone Image Acquisition
- Image Preprocessing
- Data Augmentation
- Conventional CNN Model Training
- Transfer Learning using MobileNetV2

- Transfer Learning using EfficientNetB0
- Evaluation
- Classification

Based on the identified requirements, the machine learning pipeline was designed.

4.2.2 System Architecture Design

A machine learning pipeline architecture is adopted for the proposed system. In the proposed system, the smartphone images obtained from the Sri Lankan market are saved in the custom dataset. Later, the images are preprocessed and augmented before training the models.

The architecture includes:

- Smartphone Image Collection Module
- Image Preprocessing Module
- Data Augmentation Module
- CNN Model Training Module
- Transfer Learning Module
- Evaluation Module
- Prediction Module

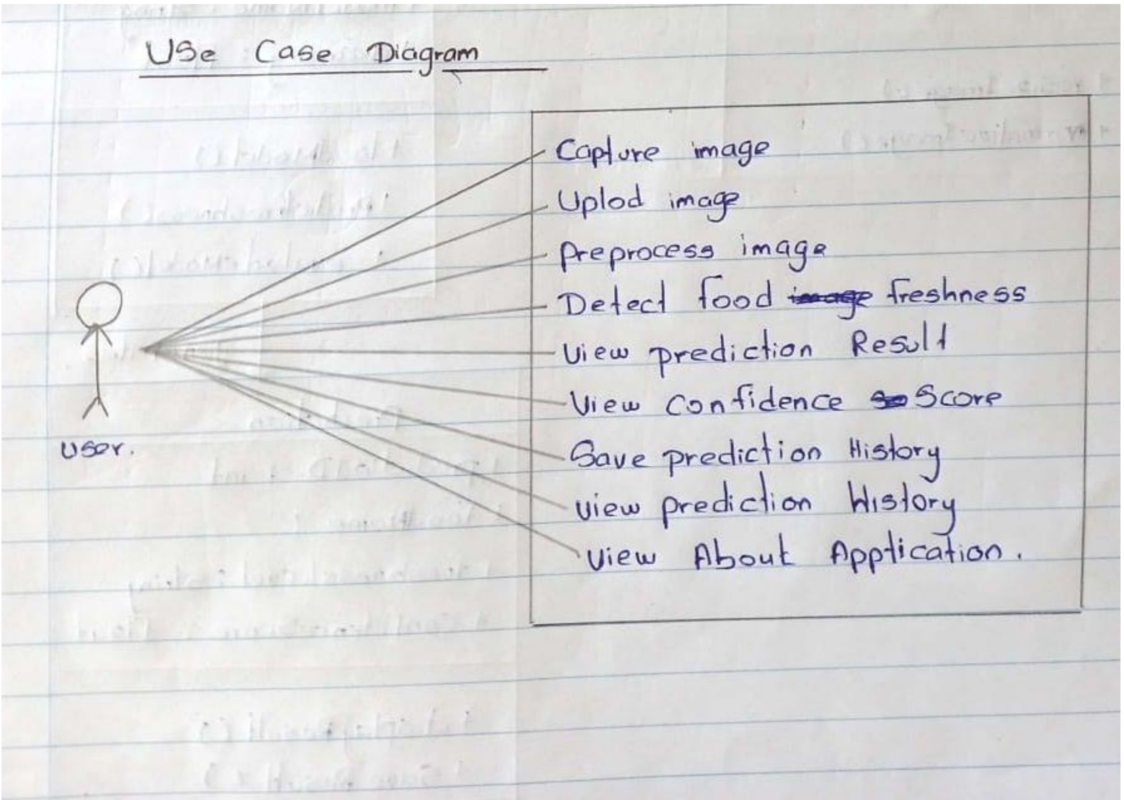
The interaction among these modules provides an efficient process of training and testing the lightweight CNN models.

The proposed architecture will allow.

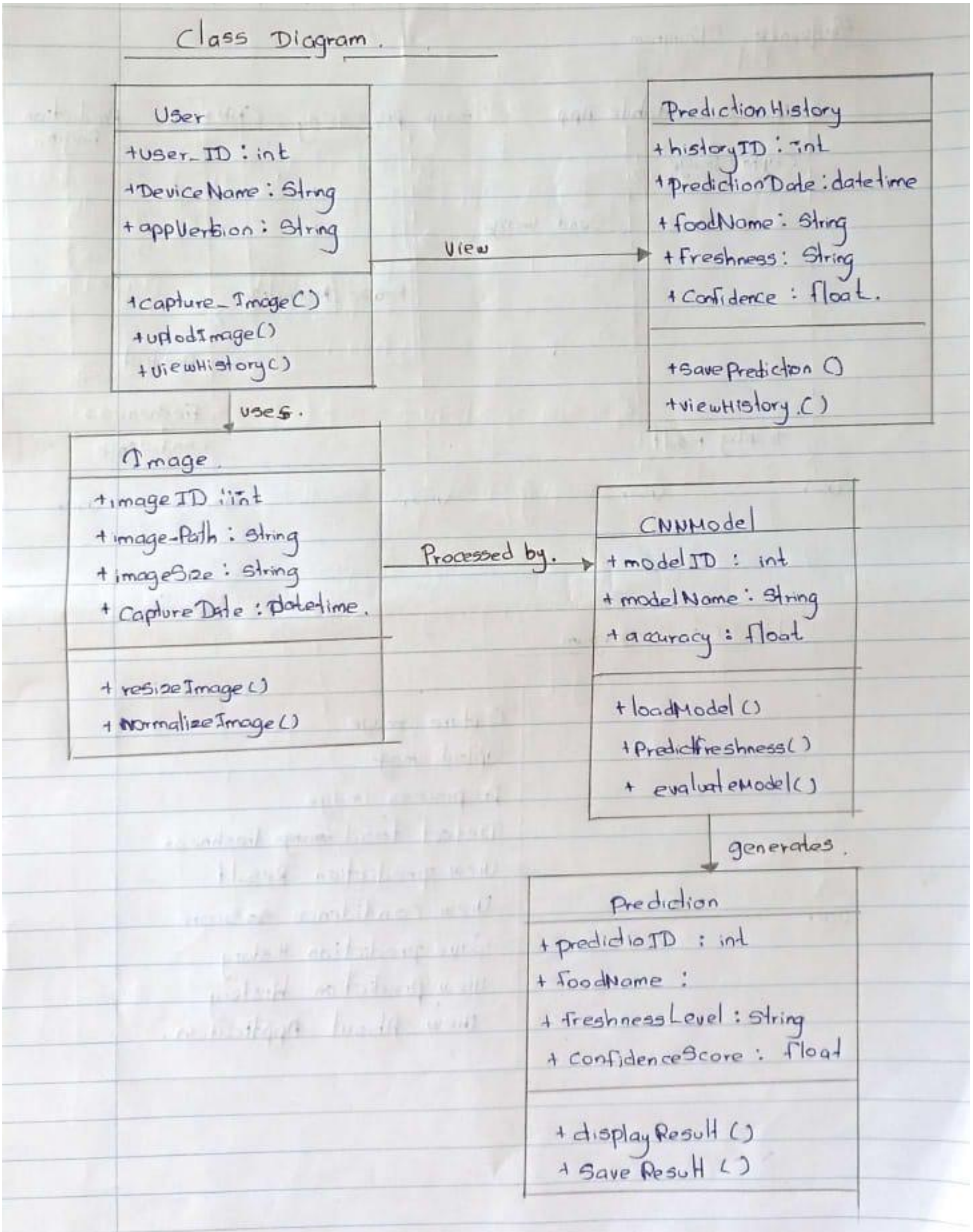
- Smartphone image acquisition,
- Automated image preprocessing,
- Image normalization,
- Image augmentation,
- Lightweight CNN training,
- Multi-class freshness classification,
- Model comparison,

- Offline prediction,
- and future mobile application deployment.

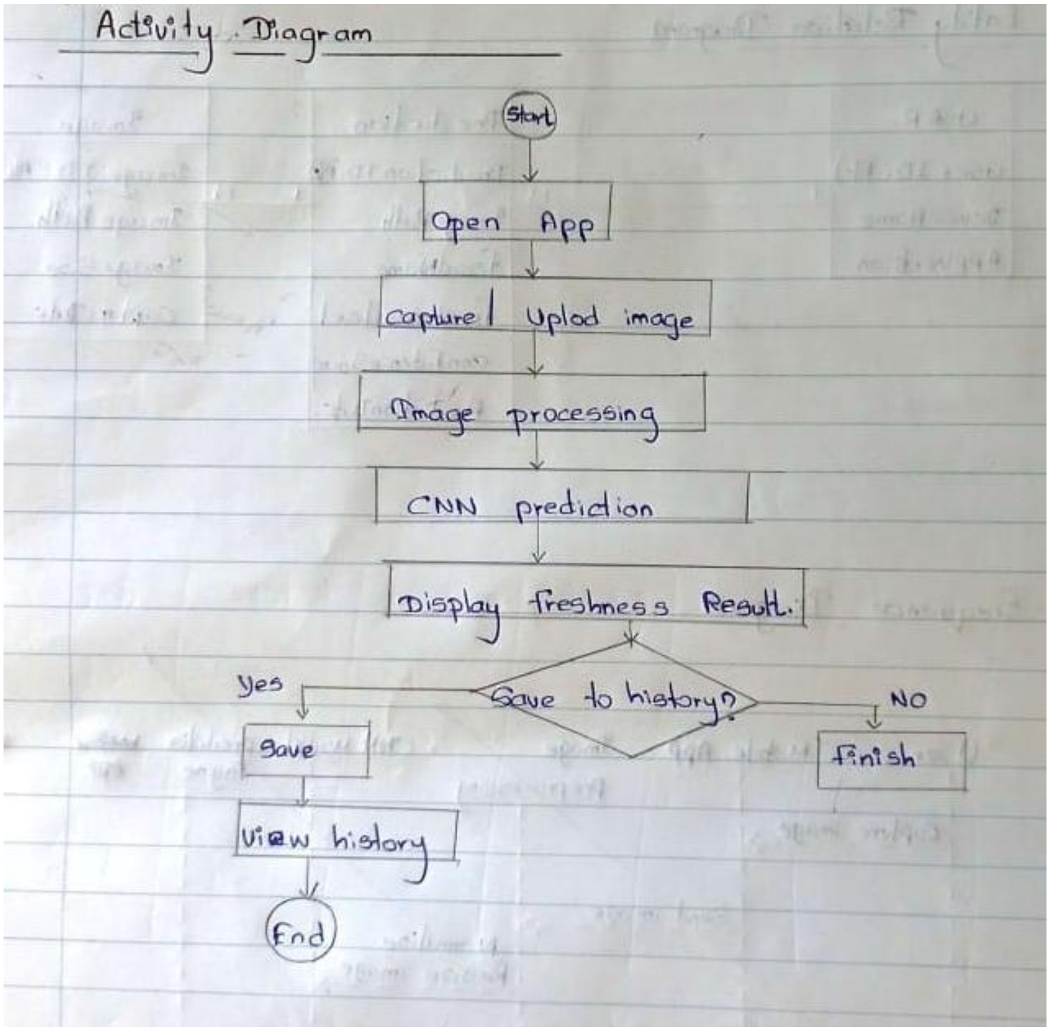
Use Case Diagram



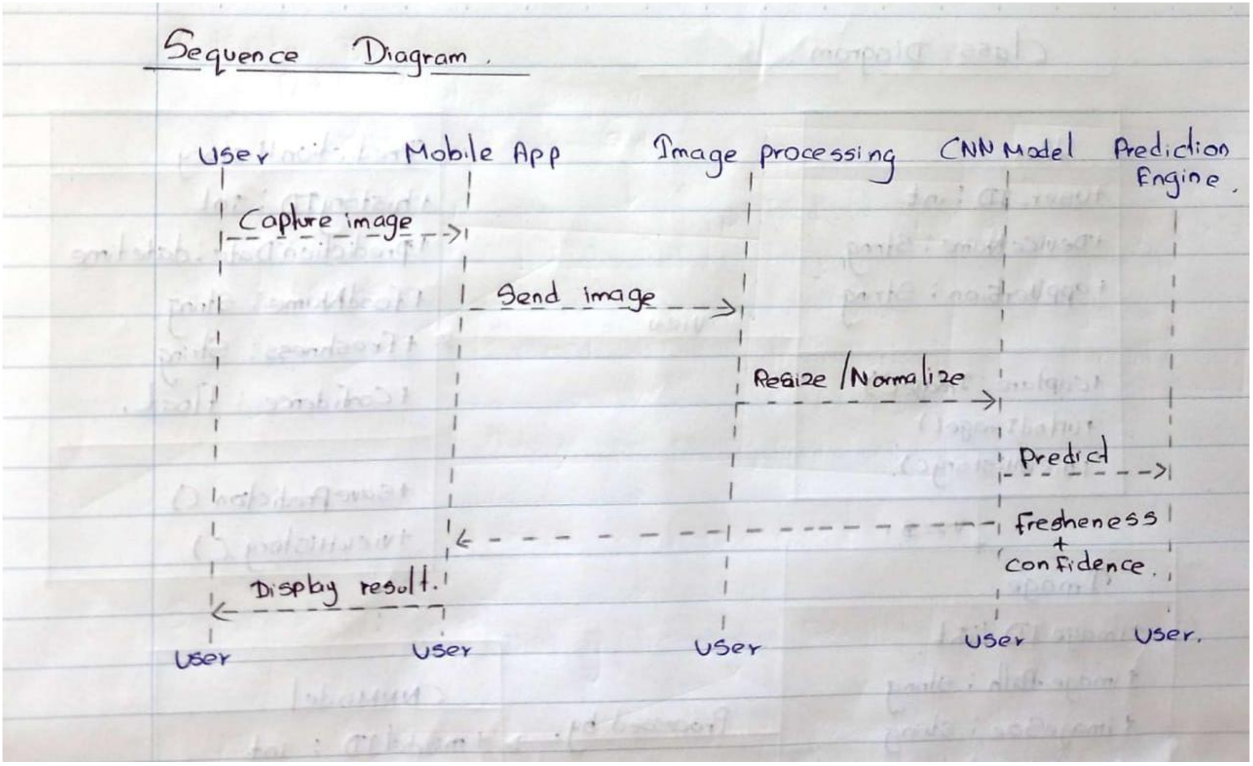
Class Diagrams



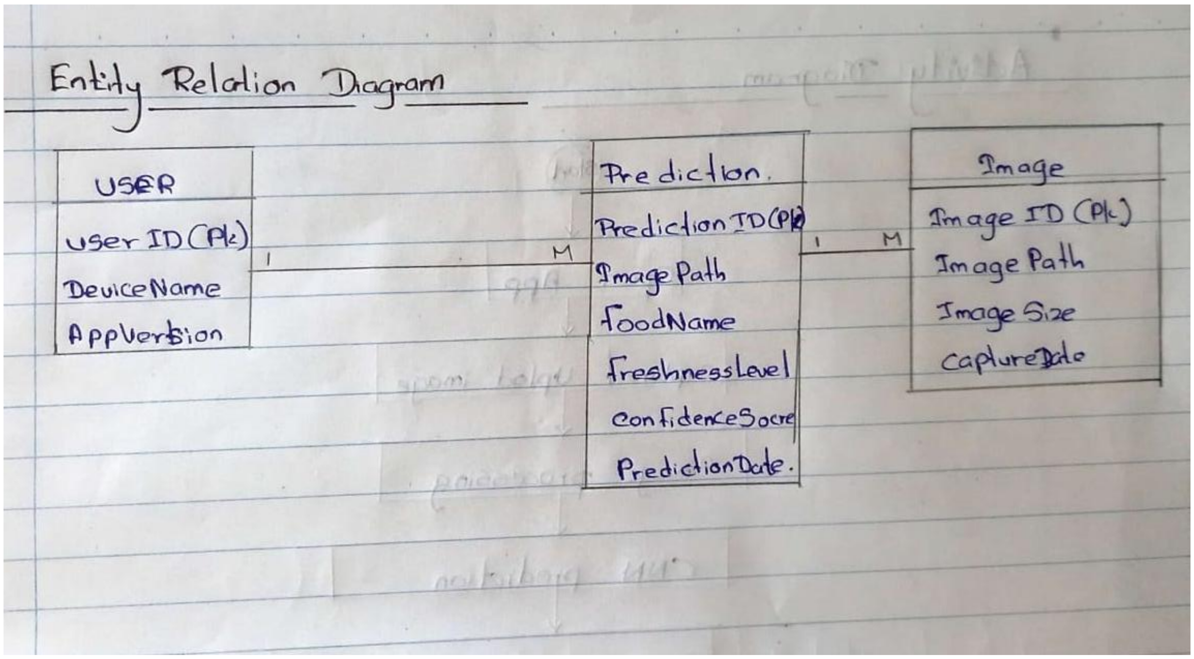
Activity Diagram



Sequence Diagrams



Entity Relation Diagram



4.2.3 Component Design

The main components of the system are:

Image Acquisition Component

- Capture images from smartphones.
- Arrange images in dataset folders.
- Classify freshness levels.

Image Preprocessing Component

- Resize images.
- Pixel normalization.
- Enhance image quality.

Data Augmentation Component

- Rotation

- Flip Horizontally
- Zoom
- Change Brightness
- Height and Width Shift

CNN Training Component

- Training Traditional CNN.
- Parameter optimization.
- Saving of trained weights.

Transfer Learning Component

- Fine tune MobileNetV2.
- Fine tune EfficientNetB0.
- Performance Comparison.

Prediction Component

- Read test image.
- Determine Freshness Level.
- Show confidence score.

Evaluation Component

- Find accuracy.
- Find Precision.
- Find Recall.
- Find F1-Score.
- Find Confusion Matrix

4.2.4 Planning the Interface Design

While the current research project is centered around the construction of the model, a basic interface for prediction will be created.

This interface will allow users to:

- Load an image from the smartphone.
- View the loaded image.
- Make a prediction regarding the freshness category.
- Display the confidence level.
- Present the results of the prediction.

Interface design is supposed to be easy, intuitive, and able to integrate into future mobile apps.

4.2.5 Validation Planning

Prior to finalizing the suggested model, the following steps will be taken for its validation:

- Checking the dataset for validity.
- Monitoring the training and validation accuracies.
- Tuning hyperparameters.
- Cross-validation.
- Comparison of performance.
- Receiving feedback from supervisors.

4.2.6 Baseline Model

A classical Convolutional Neural Network (CNN) will be used as the baseline model for the experiment. The baseline CNN model is made up of several convolution layers combined with Rectified Linear Unit (ReLU) activations, max pooling layers, a flatten layer, fully connected dense layers, and finally, the Softmax output layer.

The baseline model is a benchmark for measuring the success of the proposed transfer learning models in terms of efficiency and effectiveness. It will be compared to MobileNetV2 and EfficientNetB0 using identical datasets and evaluation metrics.

The baseline CNN model comprises the following layers:

Input Layer (224 × 224 × 3 RGB Image)
Convolutional Layers (Three)
ReLU Activation Functions
Max Pooling Layers
Flatten Layer
Fully Connected Dense Layer
Output Layer (Softmax)

The baseline CNN allows the comparison of classic deep learning models with lightweight CNN models for mobile applications.

4.2.7 Proposed Lightweight CNN Models

For achieving better classification results while keeping the computational cost low, two lightweight CNN architectures shall be deployed through transfer learning techniques.

MobileNetV2

MobileNetV2 is a type of lightweight CNN, which is primarily designed for mobile and embedded systems. This neural network is based on depthwise separable convolutions and inverted residuals blocks, enabling the reduction of a number of model parameters and at the same time ensuring high classification accuracy. Being computationally inexpensive and fast in performing operations, MobileNetV2 is very appropriate for smartphone-based application for detecting food freshness.

EfficientNetB0

EfficientNetB0 is a new optimization of the convolutional neural network, based on compound scaling, balancing depth, width and input image resolution. In contrast to the classical CNN architectures, the proposed model provides better classification results while having less number of parameters. The model is supposed to demonstrate better feature extraction performance without increasing computational cost.

Both models will be fine-tuned through transfer learning technique to classify food freshness into Fresh, Medium Fresh and Rotten categories.

4.2.8 Model Comparison

The implementation performance of the developed models will be evaluated with the use of the same training data and evaluation metrics.

Model	Functionality
Baseline Traditional CNN	Baseline Model
Lightweight MobileNetV2	Transfer Learning Model
Lightweight EfficientNetB0	Transfer Learning Model

Evaluation Metrics Accuracy, Precision, Recall, F1-Score, Efficiency, Prediction Time.

4.2.9 Experimental Setup

The suggested deep learning models shall be designed and analyzed based on the following software and hardware setup.

Parameter	Value
Programming Language	Python 3.x
Deep Learning Library	TensorFlow / Keras
Image Processing Library	OpenCV
Environment	Google Colab / Jupyter Notebook
Optimizer	Adam
Learning Rate	0.001
Batch Size	32

Epochs	30
Loss Function	Categorical Crossentropy
Image Size	224 × 224 pixels

The tests shall be conducted using the GPU support, when possible, in order to decrease training time.

4.2.9 Validation Planning

The developed models will be evaluated using both performance metrics and experiment evaluation methodology.

The dataset will be split into three sets having 70%, 15%, and 15% portions corresponding to training data, validation data, and test data. During training, the validation accuracy and loss will be considered to avoid overfitting and find the best parameters of the model.

The following metrics will be used for evaluation of the model's performance:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix
- Training Loss
- Validation Loss

The performance of the baseline CNN model, MobileNetV2 model, and EfficientNetB0 will be compared to choose the most appropriate model for lightweight food freshness detection on smartphones.

4.3 Interface Design

The interface is designed to create an environment which is simple and user friendly in order to predict the freshness of fruits and vegetables by uploading a smartphone picture of the fruit/vegetable and showing the results of prediction and confidence level.

4.3.1 Design Principles

Simplicity

Only basic functionalities are included in the interface to make sure that it will be operated easily.

Consistency

All buttons, font styles and colors are consistent all over the interface.

User Centered Design

The interface is created considering non-expert users such as consumers, vendors and researchers.

Immediate Feedback

Results are instantly displayed after processing the picture.

Accessibility

The interface includes easy-to-read font and simple controls.

4.4 Dataset Design

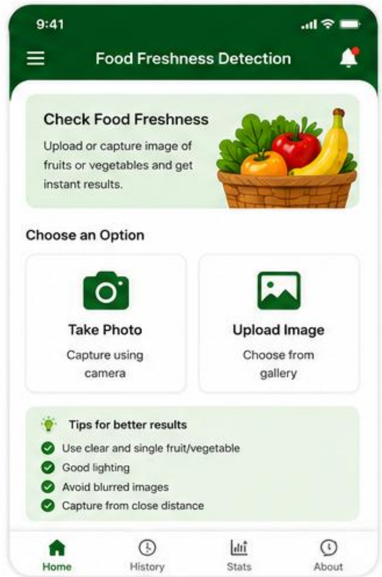
4.4.1 Dataset Structure

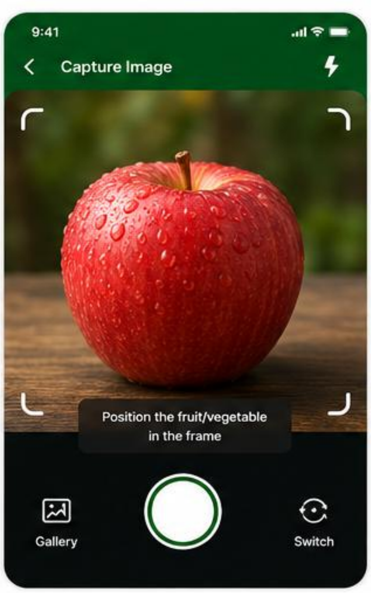
The dataset consists of pictures of fruits and vegetables taken using smartphones in Sri Lankan markets.

Pictures are classified into one of the following categories:

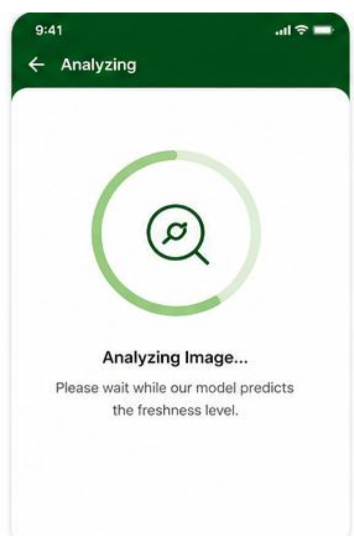
- Fresh
- Medium Fresh
- Rotten

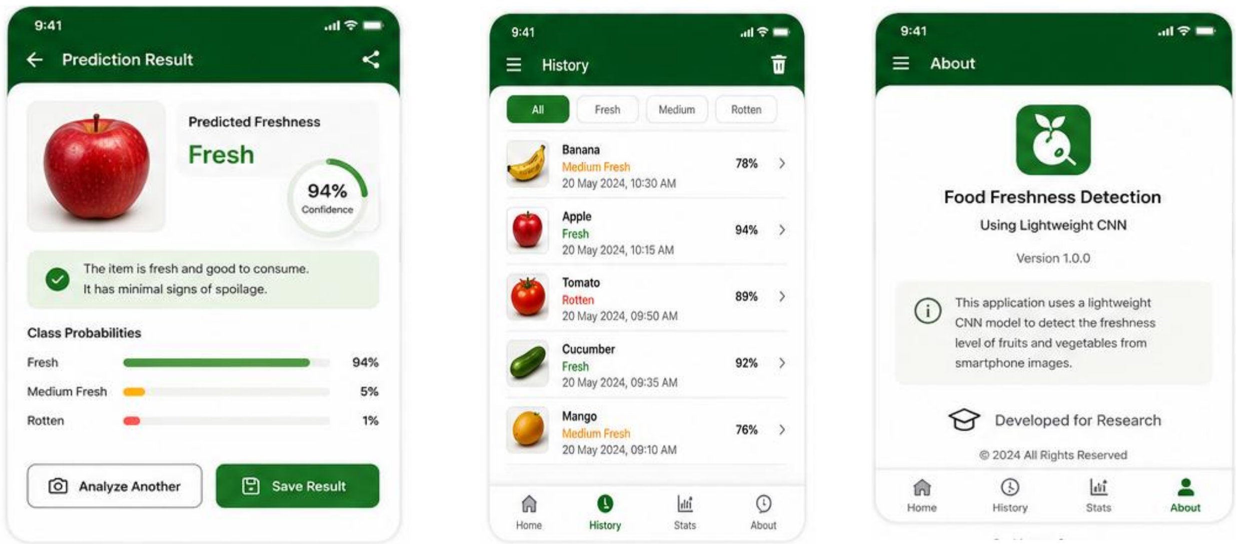

9:41
Food Freshness Detection
Using Lightweight CNN
Loading...


9:41
Food Freshness Detection
Check Food Freshness
Upload or capture image of fruits or vegetables and get instant results.
Choose an Option
Take Photo
Capture using camera
Upload Image
Choose from gallery
Tips for better results
Use clear and single fruit/vegetable
Good lighting
Avoid blurred images
Capture from close distance
Home
History
Stats
About


9:41
Capture Image
Position the fruit/vegetable in the frame
Gallery
Switch


9:41
Image Preview
Image looks good?
Retake
Analyze


9:41
Analyzing
Analyzing Image...
Please wait while our model predicts the freshness level.



4.4.2 Dataset Organization

The dataset includes:

- Training set (70%)
- Validation set (15%)
- Testing set (15%)

Images are organized under categories of their freshness cate.

4.4.3 Image Preprocessing

Preprocessing consists of the following steps:

- Image resizing (224×224 pixels)
- Pixel normalization
- Augmentation
- Image labeling

These processes help improve the performance of the model and prevent overfitting.

4.5 Hardware and Software Requirements

4.5.1 Smartphone Camera

A smartphone camera is used to take images under real-life market conditions of Sri Lanka.

4.5.2 Software Environment

Software environment used for developing the proposed solution include:

- Python
- TensorFlow
- Keras
- OpenCV
- NumPy
- Matplotlib
- Google Colab / Jupyter Notebook

4.5.3 Hardware Requirements

For the training process, hardware requirements are:

- Intel Core i5 (or equivalent) processor
- At least 8 GB RAM
- GPU (optional for faster training process)
- Smartphone for capturing images

Git Repository

https://github.com/SHANALIKA/Food-Freshness-Detection-System.git